

DAY 2 OF 30

Backend Foundation

Express Setup

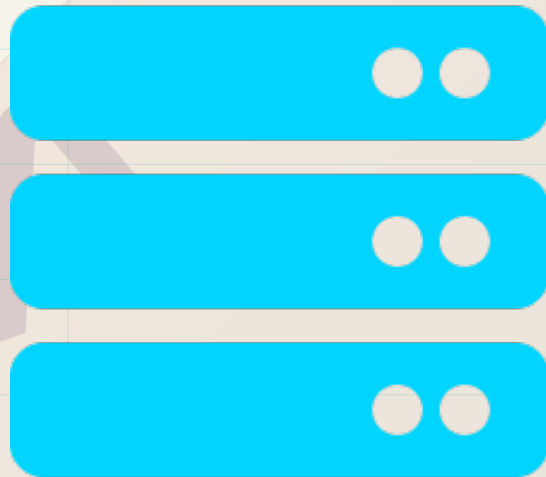
Building Our First Backend Application

Node.js

Express

MongoDB

REST API





What Will We Learn Today?



1



Create Express Server

Setup Node.js + Express

2



Create API Routes

GET, POST, PUT,
DELETE

3



Add Middleware

Request processing
pipeline

4



Connect MongoDB Atlas

Cloud database
connection

5



Test APIs with Postman

API testing & debugging

"Today we build the brain of our application" 🧠



Required Software



Node.js

Run JS Outside Browser



MongoDB Atlas

Store Data Online



VS Code

Write Code



Git

Track Changes



GitHub

Store Code Online

Simple Analogies



VS Code

= Writing Book



MongoDB

= Store Room



Backend

= Delivery System



GitHub

= Online Library



How HomelyHub Works — 3-Tier Architecture



Frontend

What User Sees

Login · Property List · Booking



React Frontend



Backend

Application Brain

Validate User · Process Request · Business Logic



Node + Express Backend



Database

Application Memory

Users · Properties · Bookings



MongoDB Database



React.js

JavaScript UI Library

Components · State · Routing



Express.js

Node.js Framework

REST APIs · Middleware · Routes



MongoDB

NoSQL Cloud Database

Collections · Documents · Atlas



Complete Request Journey



 **User Clicks Button**



 **React Frontend**



 **Express Backend**



 **MongoDB Database**



Real World Example — Book a Stay

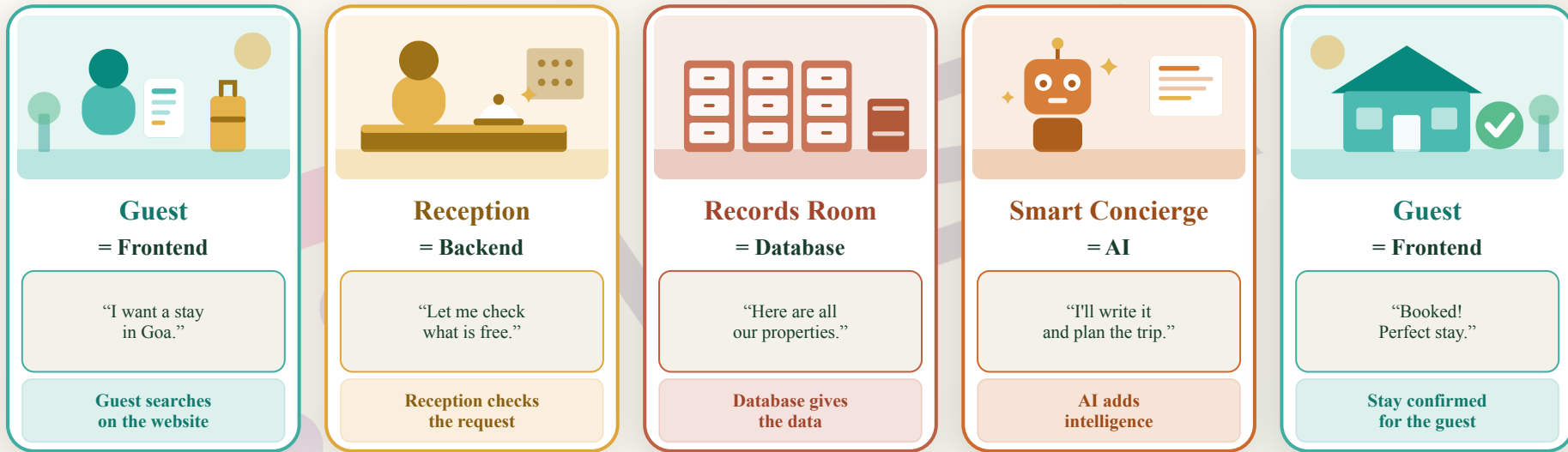
-  Guest Clicks Book Now
-  React Sends HTTP Request
-  Backend Verifies User Token
-  Backend Saves Booking to DB
-  MongoDB Stores Data
-  Response Returns to React
-  Booking Confirmed 



Let's Understand Our App Architecture!



Think of it like a hotel 🏨



Guest asks → Reception checks → Records room provides → AI adds magic → Stay confirmed

✨ Just like a great hotel, every part works together to give an amazing stay! 🏡



Project Initialization



HomelyHub



backend



Node.js

JavaScript Engine



npm

Developer App Store



init

Create New Project



-y Flag

Accept All Defaults



Terminal

```
$ cd HomelyHub/backend
```

```
$ npm init -y
```

```
✓ package.json created!
```



package.json — Heart of Every Node.js Project



Project Name



**package
.json**



Dependencies

⚙ Run Scripts



Version Info

```
{ "name": "homelyhub-backend", "version": "1.0.0", "scripts": { "dev": "nodemon src/index.js" },
```

```
"dependencies": { "express": "^5.1.0", "mongoose": "^8.15.2", "dotenv": "^16.5.0", "bcrypt": "^6.0.0" } }
```



Why package.json Matters — Team Collaboration Story



Developer A



Developer A writes code



npm install express



package.json saves dependency



Pushes to GitHub



Developer B



Developer B joins team



git clone project



npm install



Everything works!



Quick Revision — Key Concepts



Frontend

Property Pages



Backend

Booking Logic



Database

Users & Bookings



Node.js

JavaScript Runtime



npm

Package Manager



package.json

Project Heart



Ready To Build Our First Express Server?

⚡ Let's Start Coding ⚡



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